Research Project Periodic log entry form

Project ID: 25-26J-189

Student ID: IT22309860

Name: Rasingolla W.M.S.N.K

Individual Component: Pollination Readiness (Detection)

Summary of RP work carried out during the reporting period.

Date	RP work.
12/01/2025	Reviewed existing IoT based smart agriculture systems. Featured gaps identified.
12/02/2025	Studied CNN and Yolo based flower detection techniques. Selected Yolo architecture for the real time detection
12/03/2025	Selected sensors (DHT22, BH1750, soil moisture) for environmental monitoring. sensor list finalized with accuracy justification.
12/04/2025	Prepared data set requirements for flower image collection. Data imbalance risk identified.
12/05/2025	Designed mobile application workflow for farmers. Userflow optimized for field usability.
12/06/2025	Reviewed expert feedback expectations & validation metrics.
12/07/2025	Designed database schema for sensor and image data ~~fch~~(Scalable schema).

Data	RP work.
~~08/12/2025~~ 12/08/2025	Reviewed blockchain-based data security concepts Decided on secure cloud storage approach.
~~08~~ 12/09/2025	Analysed pollination scheduling algorithms. Rule-based + ML hybrid approach selected.
12/10/2025	Documented limitations of manual pollination practices.
12/11/2025	Economic benefits referenced. Reviewed 3 case studies on smart farming profitability.

Supervisor's comments.

Student progressing well in research Project :
Yes / No.

If No;
project work to be carried out before next evaluation period

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Good progress Work is satisfactory and aligned with research objectives.

code based checked : Yes / No.

Signature :

Date : 13/12/2025